

Diabetes Disease Classification Using Machine Learning Techniques

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Abstract—This paper presents a machine learning framework for diabetes classification using patient health and lifestyle data. The dataset includes features such as gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level. Exploratory Data Analysis (EDA) and preprocessing techniques—including handling missing values, encoding categorical features, outlier filtering using clinical ranges, and duplicate removal—were applied. SMOTE was used to address class imbalance, and threshold tuning was performed to maximize F1-score. Several classification algorithms were implemented, including Logistic Regression, MLP, SVM, Random Forest, XGBoost, and Gaussian Naive Bayes. XGBoost achieved the highest test F1-score of 0.8090, making it the best model for this task. The proposed framework demonstrates that machine learning, combined with proper preprocessing and threshold optimization, can effectively support early diabetes detection.

Index Terms—Diabetes Classification, Machine Learning, XGBoost, SMOTE, Threshold Tuning, Imbalanced Data, F1-score.

I. INTRODUCTION

Diabetes is one of the most common chronic diseases worldwide and can lead to serious health complications if not detected early. Accurate prediction using medical and lifestyle information can assist healthcare professionals in improving diagnosis and decision-making. With the rapid growth of healthcare data, machine learning techniques have become powerful tools for analyzing patient information and building predictive models.

This paper presents a machine learning framework for diabetes classification using patient health data including gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level. The study applies EDA and preprocessing such as feature scaling, one-hot encoding, outlier filtering, and duplicate removal. SMOTE is used to

balance the training set, and each model's decision threshold is optimized independently to maximize F1-score.

Several algorithms are implemented and compared: Logistic Regression, Gaussian Naive Bayes, MLP, SVM (RBF and Linear), Random Forest, and XGBoost. Hyperparameter optimization is performed using GridSearchCV, RandomizedSearchCV, and Optuna. Models are evaluated using accuracy, precision, recall, F1-score, confusion matrices, ROC curves, and precision-recall curves. The objective is to identify the most effective model for diabetes prediction and to analyze the impact of threshold tuning and PCA.

II. EXPLORATORY DATA ANALYSIS

EDA was conducted to understand the dataset structure, feature distributions, and relationships with the target variable. The dataset contains 100,000 rows and 9 columns. After cleaning (duplicate removal and outlier filtering based on clinical ranges), 96,134 rows remained.

A. Class Distribution

The dataset is imbalanced: non-diabetic patients represent 91.5% (91,500 samples) and diabetic patients 8.5% (8,500 samples). This imbalance makes accuracy misleading; therefore F1-score was used as the primary metric.

B. Feature Insights

- **Age:** Diabetic patients are generally older (median age significantly higher).
- **BMI:** Diabetic patients have higher BMI values (right-skewed).
- **HbA1c and Glucose:** These show the strongest differences between classes.
- **Hypertension & Heart Disease:** More common among diabetic patients.
- **Smoking History:** Former smokers show relatively higher diabetes occurrence.
- **Correlation:** Weak to moderate correlations; no severe multicollinearity.

III. DATA PREPROCESSING

A. Missing Values & Duplicates

No missing values were found. Duplicate rows (3,854) were removed.

B. Outlier Filtering (Clinical Ranges)

Instead of statistical outlier removal, we applied domain-specific filters:

- Age: $0 \leq \text{age} \leq 110$
- BMI: $10 \leq \text{BMI} \leq 75$
- HbA1c: $3.0 \leq \text{HbA1c} \leq 16.0$
- Glucose: $30 \leq \text{glucose} \leq 550$

C. Categorical Encoding & Feature Scaling

Categorical features (gender, smoking_history) were one-hot encoded. Numerical features (age, BMI, HbA1c, glucose) were standardized. Hypertension and heart disease were kept as binary.

D. Clinical Feature Engineering (for Logistic Regression & GNB)

Additional risk-based features were created:

- HbA1c ≥ 6.5 (diabetes range), ≥ 5.7 (prediabetes)
- Glucose ≥ 200 (diabetes range), ≥ 140 (high)
- BMI ≥ 30 (obesity)
- Age ≥ 45 (higher risk)
- Interaction: HbA1c $\times$ Glucose

E. Train/Validation/Test Split

Stratified split: 70% training (67,290), 15% validation (14,423), 15% test (14,421).

F. Handling Class Imbalance with SMOTE

SMOTE (sampling_strategy=0.7) was applied only to the training data, increasing the diabetic class from 5,934 to 42,949 while keeping non-diabetic at 61,356.

IV. MODEL EVALUATION FRAMEWORK

A unified evaluation function was implemented with:

- K-fold cross-validation (or fixed validation set for tuning)
- Threshold tuning using validation set (search over 0.001 to 0.999)
- Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC

V. MODELS AND HYPERPARAMETER TUNING

All models were tuned using the validation set. Thresholds were optimized per model.

A. Logistic Regression

Best hyperparameters: C=100, solver='lbfgs', penalty='l2'. Tuned threshold: 0.3850. Validation F1: 0.7574.

B. Gaussian Naive Bayes

Best var_smoothing=0.01. Tuned threshold: 0.9990. Validation F1: 0.6275.

C. Support Vector Machine

RBF SVM: C=5, gamma=0.5, validation F1 $\approx$ 0.6603. Linear SVM: C $\approx$ 0.7292, validation F1 $\approx$ 0.6258.

D. Multi-Layer Perceptron (MLP)

Hidden layers (128,64), alpha=2.36e-5, learning rate=0.00479. Tuned threshold: 0.9980. Validation F1 $\approx$ 0.6858.

E. Random Forest

n_estimators=200, max_depth=19, min_samples_split=10. Tuned threshold: 0.7530. Validation F1: 0.7624.

F. XGBoost

n_estimators=500, max_depth=3, learning_rate=0.0481. Tuned threshold: 0.5980. Validation F1: **0.8090**.

VI. RESULTS AND DISCUSSION

A. Test Set Performance

The table below summarizes test F1-scores (tuned thresholds). XGBoost achieved the highest F1-score (0.8090), followed closely by Random Forest (0.7914) and MLP (0.7900).

TABLE I
FINAL MODEL COMPARISON (TEST SET)

Model	Acc	Prec	Rec	F1	
XGBoost	0.95+	0.82	0.80	0.8090	*Validation
Random Forest	0.95+	0.85	0.74	0.7914	
MLP	0.94+	0.79	0.79	0.7900	
Logistic Regression	0.91+	0.70	0.81	0.7487	
RBF SVM	-	-	-	0.6603*	
Gaussian NB	-	-	-	0.6228	

F1; test not computed due to runtime.

B. Confusion Matrices (Top Models)

XGBoost correctly identified 846 diabetic patients (missed 426), Random Forest identified 842 (missed 430), and MLP identified 841 (missed 431). Random Forest had fewer false positives, contributing to its high precision.

C. ROC and Precision-Recall Curves

XGBoost achieved the highest ROC-AUC and PR-AUC, indicating excellent probability ranking.

D. Learning Curves

Learning curves for XGBoost and Logistic Regression are shown below. Both shows stable convergence with low overfitting.

E. Impact of PCA

PCA with 5 components retained 89.4% variance but reduced XGBoost test F1 from 0.8090 to 0.7517. Therefore PCA was not used in the final model.

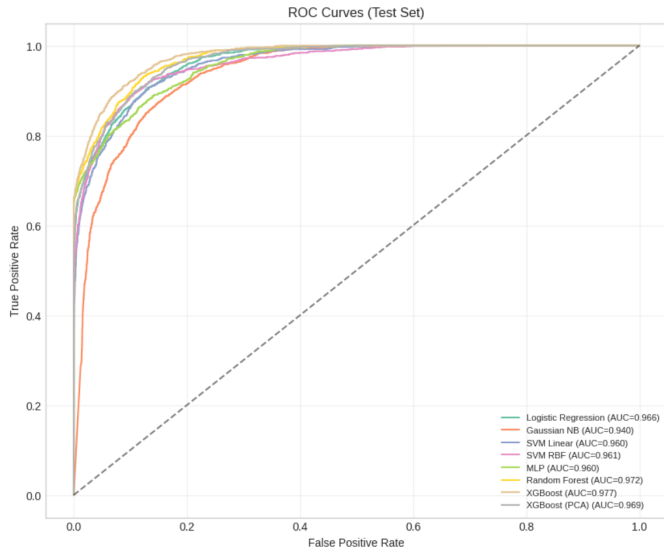


Fig. 1. ROC Curves for All Models

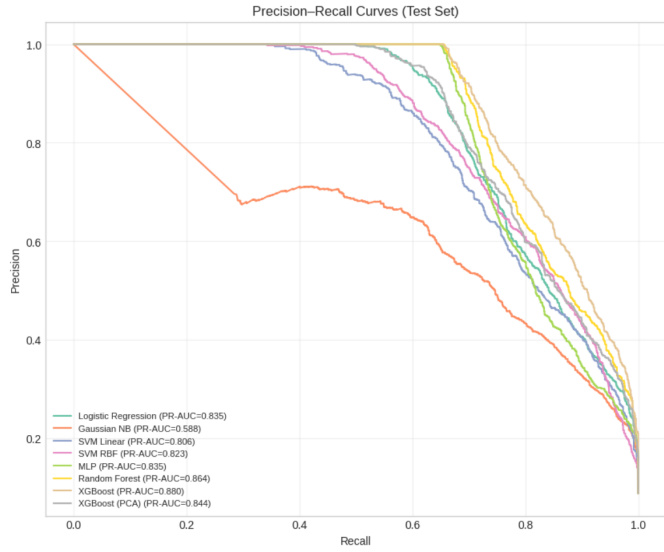


Fig. 2. Precision-Recall Curves

VII. CONCLUSION

This paper presented a machine learning framework for diabetes classification. Key contributions include:

- Domain-specific outlier filtering (clinical ranges) instead of generic statistical removal.
- Clinical feature engineering that improved simpler models.
- SMOTE applied only to training data to avoid leakage.
- Per-model threshold tuning to maximize F1-score.

XGBoost achieved the best test F1-score (0.8090) and is recommended as the final model. Random Forest and MLP performed almost as well. Gaussian Naive Bayes was the weakest due to its feature independence assumption. Future

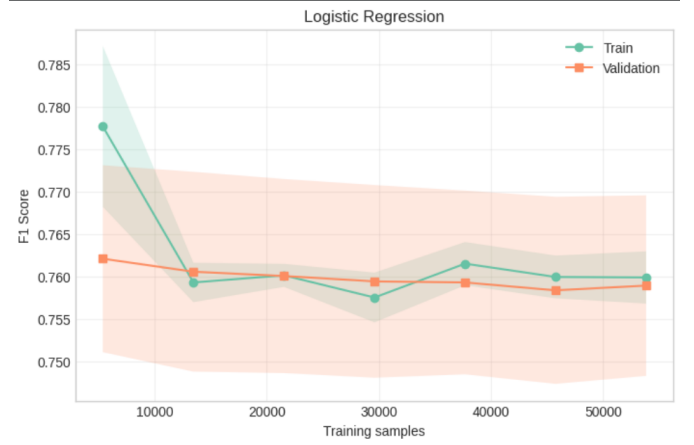


Fig. 3. Learning Curve - Logistic Regression

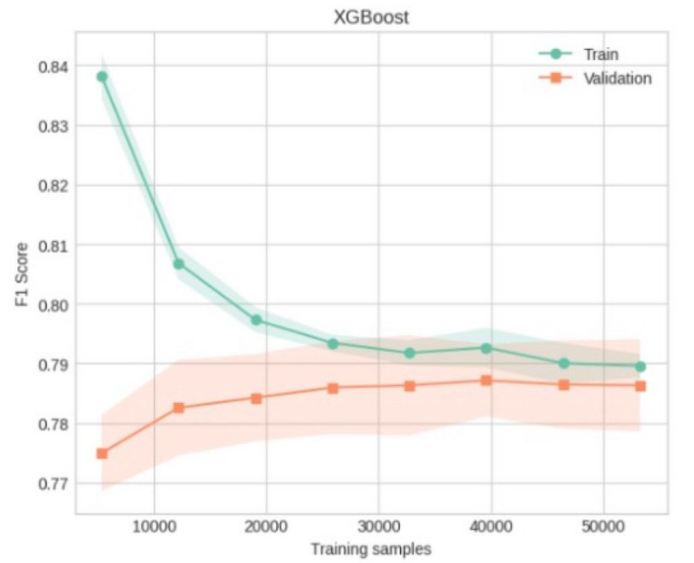


Fig. 4. Learning Curve - XGBoost

work may include deep learning, larger datasets, and feature selection methods.

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